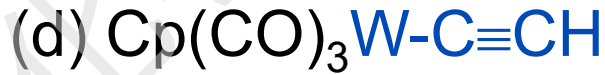
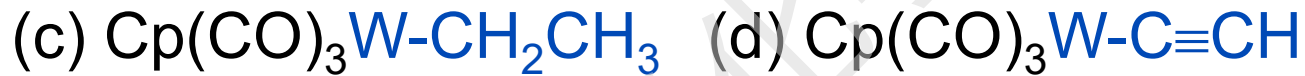




HOMEWORK

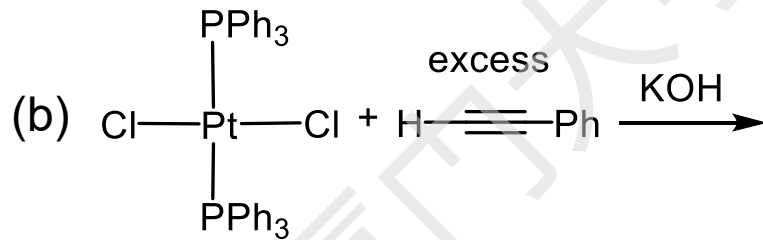
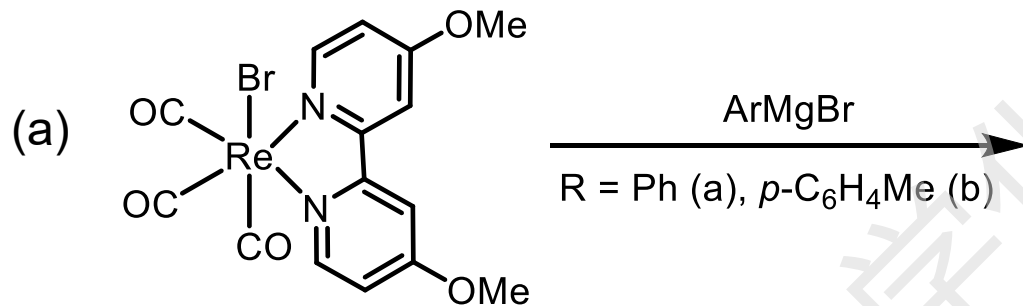
1. Which of the following complexes would you expect to have the strongest metal-alkyl bond?





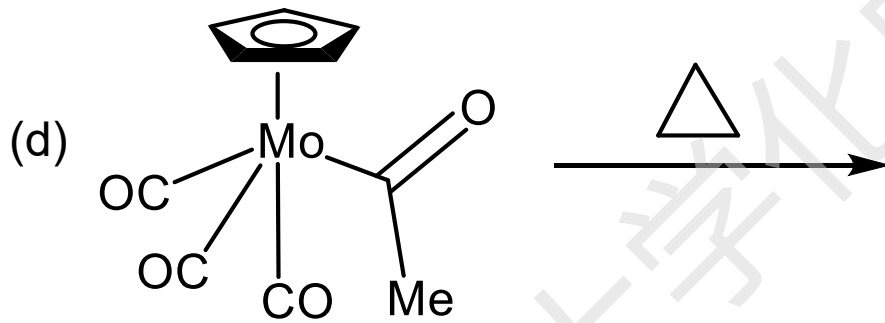
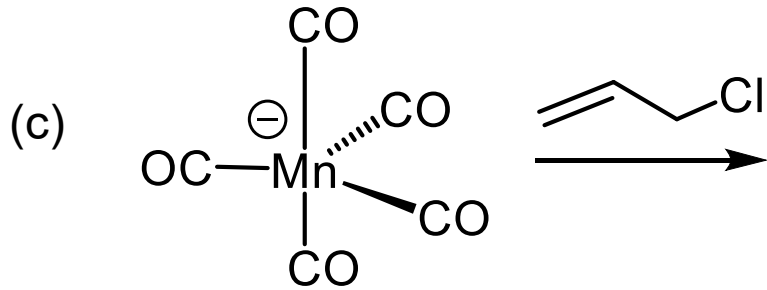
HOMEWORK

2. Predict the metal-containing products of the following reactions





HOMEWORK

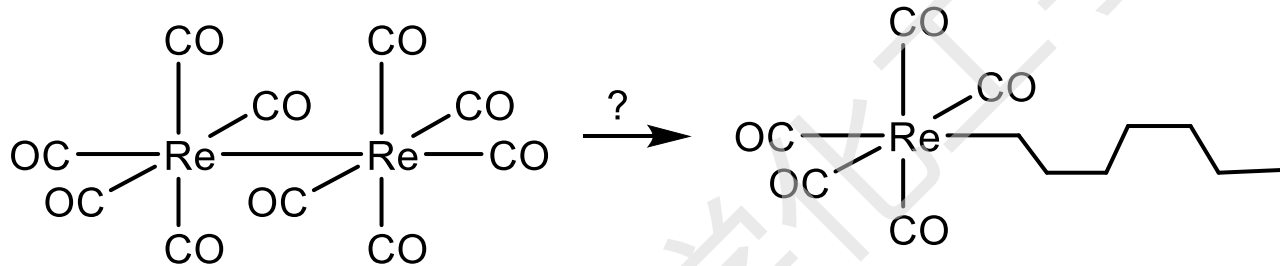




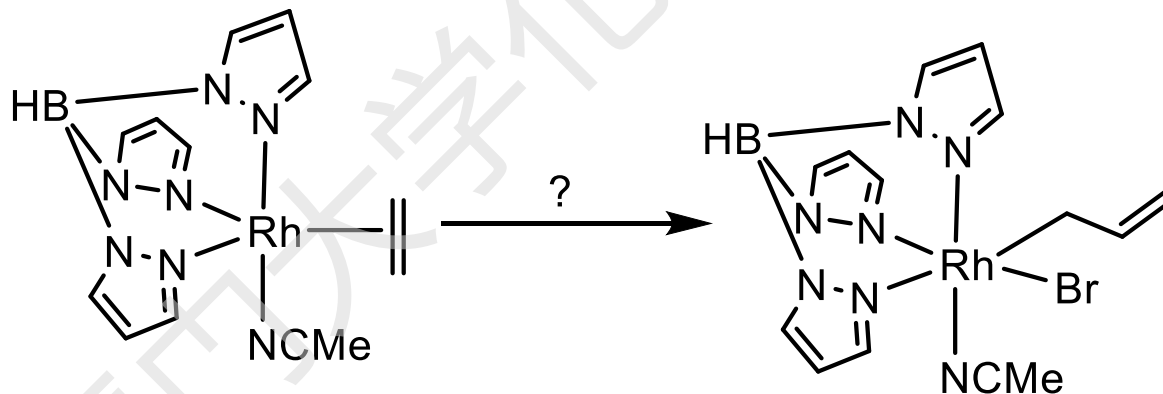
HOMEWORK

3. Provide appropriate reagents for the following transformations

(a)



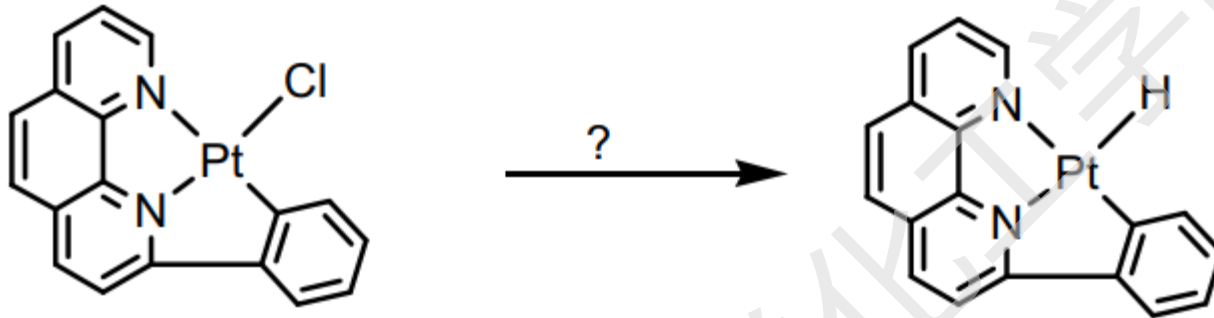
(b)



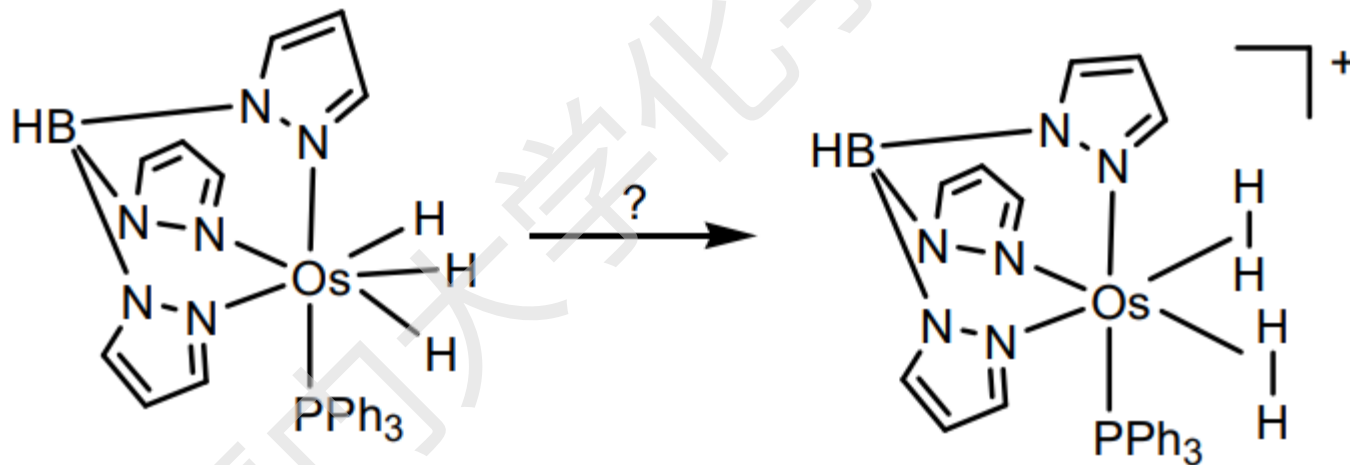


HOMEWORK

(c)



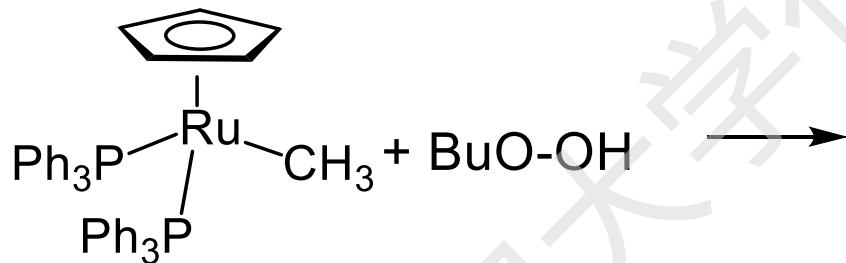
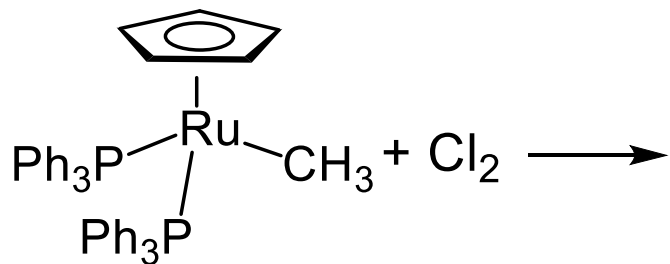
(d)





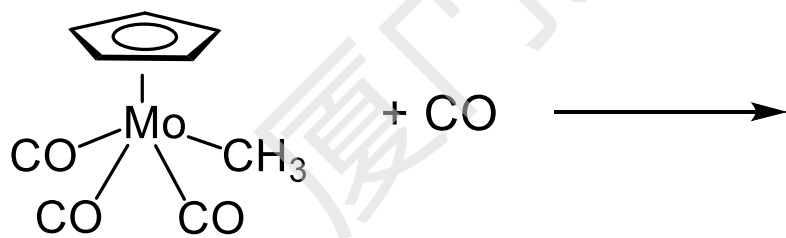
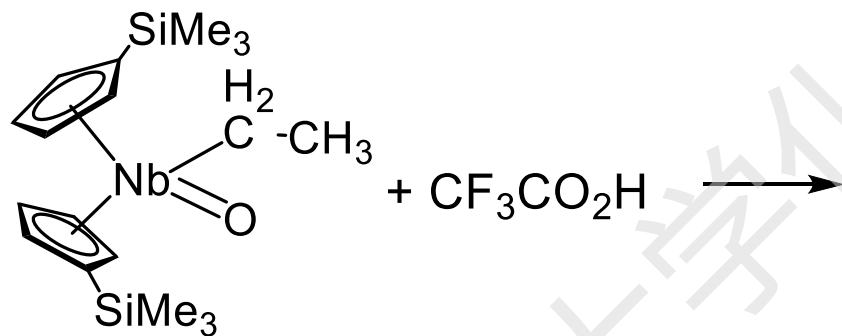
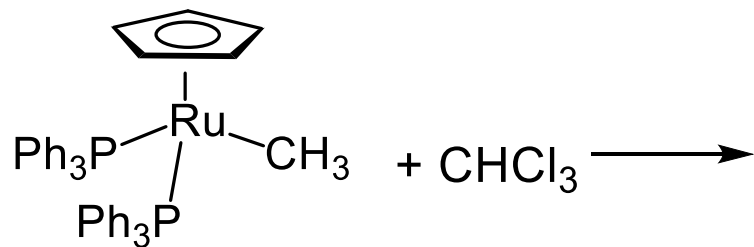
HOMEWORK

4. Predict the metal-containing products of the following reactions





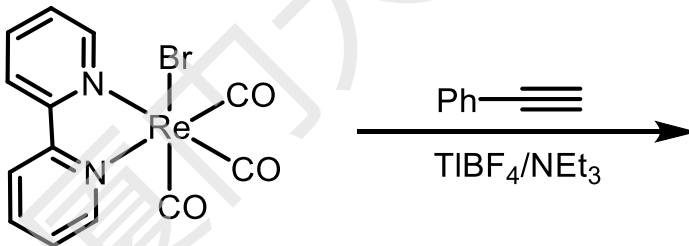
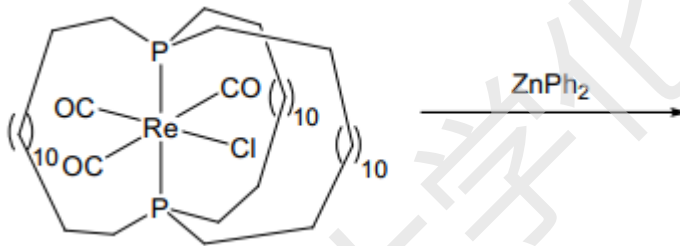
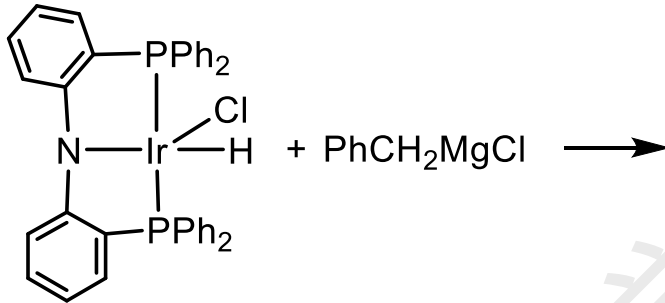
HOMEWORK





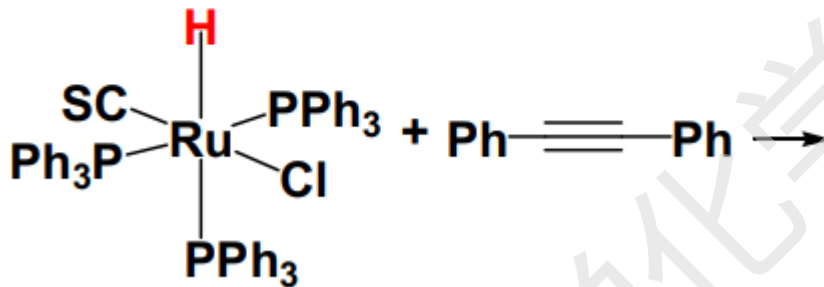
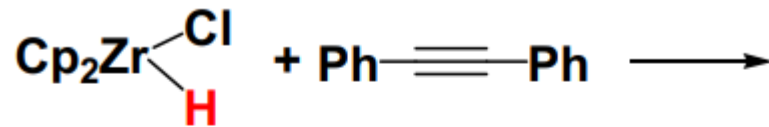
HOMEWORK

5. Predict the metal-containing products of the following reactions





HOMEWORK





Bonus Challenge:

Catalytic *cis*-semihydrogenation of internal alkynes is one of the most important approaches for synthesis of Z-alkenes.

There are two known pathways for the transfer hydrogenation (TH) of C–C triple bonds with alcohols: an inner-sphere dihydride mechanism and outer-sphere metal–ligand bifunctional catalysis. **Please provide schematic illustration of these two mechanisms for TH of alkyne with alcohol.**